



Accelerating the discovery of battery electrode materials through data mining and deep learning models

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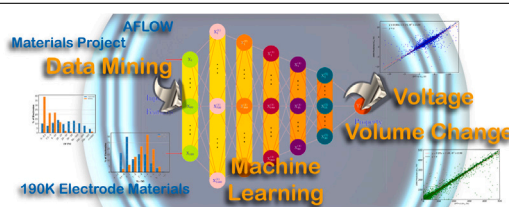
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HIGHLIGHTS

- A new and robust database for battery electrode materials is built.
- A set of potential new electrode materials is identified from the new database.
- ML models built using the new database show improvement compared to previous models.

GRAPHICAL ABSTRACT



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ABSTRACT

The availability of crystalline materials databases allows for building accurate machine learning (ML) models that can accelerate the exploration of materials chemical space for energy storage applications. In this work, we screen all inorganic materials included in the Materials Project and AFLOW databases as potential metal-ion battery electrodes. We develop an efficient protocol to mine and screen raw data in current databases and provide a new database of electrode materials by considering pairs of charged and discharged electrodes. This effort leads to a new database with over 190,000 instances, in contrast to the original battery database which contains about 5000. **The expanded battery data set is then used to build regression-based deep neural network models for predicting average voltages and percentage volume changes upon charging and discharging, which present improvements of at least 28% for target properties with respect to previous models, and are now able to predict anode electrodes (low voltage region) as well as electrodes that will not work in electrochemical cells (negative voltages), overcoming the challenges identified in previous ML models for battery electrodes.** Additionally, a further screening of the expanded database itself allowed us to identify 35 novel electrode candidates with excellent battery performance metrics.

1. Introduction

The huge demand of materials for renewable energy entails fast and efficient ways of exploring materials' space in search of best candidates for targeted applications. Energy storage materials are of particular interest in the scientific community as the world races to develop electric vehicles and transportation systems to replace carbon-emitting technologies [1,2]. Additionally, consumer electronics and power grids also require the use of efficient and long life-cycle energy storage systems [3–5].

Over the last decade, researchers have developed a number of databases of materials for multiple purposes. Some of these databases contain materials that were experimentally synthesized such as for instance, the Inorganic Crystal Structure Database (ICSD) [6], while others contain materials that have been computationally modeled through high throughput quantum mechanical methods, such as the Materials Project (MP) [7,8], AFLOW [9], Open Quantum Materials Database (OQMD) [10], and Novel Materials Discovery (NOMAD) [11] databases. **Despite the large number of data instances in these databases, only a limited number of those materials have relevant**

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information reported for their potential utilization in energy storage applications. MP, for instance, includes 144,595 inorganic compounds, out of which only about 5,000 have intercalation battery electrode properties computed.

Machine learning (ML) algorithms have become indispensable tools for accelerating the exploration of materials' physical and chemical space for potential electrodes beyond the more traditional experimental and quantum mechanical methods for studying battery electrode materials [12–16]. ML methods are particularly attractive due to the minimal time and computational resources that they require, in contrast to experimental and quantum mechanical methods. Besides battery electrodes, ML models have also been developed for a number of energy storage and conversion studies [17] including solar cells [18, 19], catalysis [20,21], and battery electrolytes [22–28]. However, the availability of data is crucial to the development of robust ML models. Supervised ML models are trained by learning to evaluate a given property with the use of input parameters known as features. The predictive power of the models depends largely on the quality and quantity of the data used in training these models [29].

Taking advantage of the availability of crystal data in current databases, our present work focuses on data mining potential battery electrodes from the large raw data set of crystalline materials in MP and AFLOW. The data mining is carried out by developing an effective algorithm that searches for pairs of materials that can potentially serve as charged and discharged metal-ion electrodes. The main properties of interest, namely the average voltage, percentage volume change, specific capacity, and specific energy are then evaluated for the identified pairs. From this new database, a subset of novel electrodes with excellent performance metrics for rechargeable battery applications are identified. Additionally, the robust data set is also used to develop deep learning models for accurate predictions of electrode properties.

2. Methods

2.1. Battery data mining

The sources of our data for this work are the MP and AFLOW databases. AFLUX [30] was used to query the materials and their properties from the AFLOW database while, for the MP database, we downloaded the crystalline materials identification number (ID) using the Materials Application Programming Interface (MAPI) [31] and the Python Materials Genomics (pymatgen) [32]. For each of the material's ID in MP, the structure, along with relevant properties were then queried. These properties include the space group, the number of atoms in a unit cell, the atomic species in the unit cell, the unit cell composition, and the DFT+U computed energy and volume of the unit cells. From the resulting data, we identified and removed non-solid structures since our interest is in solid state electrode materials. Duplicates were also removed from the data. After this initial processing of the data, the databases resulted in 124,397 crystalline solid materials for the MP database and 76,031 instances for AFLOW.

We then embarked on the search for pairs of potential discharged and charged electrodes from the lists of materials from the MP and AFLOW, respectively. The search is focused on electrodes for active metal ions M ($M = \text{Li, Na, K, Rb, Cs, Mg, Ca, Al, Zn or Y}$). A flow chart of the algorithm used for the search is shown in Fig. 1. The unit cell of material N in the list with index i ($i = 0, 1, 2, \dots, l-1$, where l is the last index) is compared to the unit cell of material N^* with index $i+1, i+2, i+3, \dots, l$. N and N^* are passed on to the next step if the atomic species in N and N^* are the same or differ only by M (the active metal). Then we check the ratio of the number of atomic species in N and N^* , respectively. If the ratio of the number of a particular atomic type in N is the same as that of the same atom in N^* for all the different atomic species present (in both N and N^*), but is different for the active metal M , then N and N^* are passed on to the next step. To ensure that N and N^* are commensurate, and in order

to know the actual difference in the number of M in N and N^* , the ratio of the number of atom type A (n_A) in N to the number of the same atom type A^* (n_{A^*}) in N^* is obtained. If the ratio $\frac{n_A}{n_{A^*}}$ is greater than one, the number of different atom types in N^* is multiplied by $\frac{n_A}{n_{A^*}}$, otherwise the number of different atom type in N is multiplied by $\frac{n_{A^*}}{n_A}$. The same factor, $\frac{n_A}{n_{A^*}}$ or $\frac{n_{A^*}}{n_A}$, whichever is greater than one, is then used to multiply both the energy and volume per unit cells of N^* or N , respectively. The energy of the M is also modified accordingly. After the modification, if the number of the M (for instance, the number of Li atoms) in N is greater than that of the N^* , then N is considered the discharged and N^* the charged electrode; otherwise, N^* is the discharged and N is the charged electrode.

Using the DFT+U energies and volumes from the MP and AFLOW databases, the average voltage (V_{av}), the percentage volume change ($\Delta V_{\%}$), the specific capacity (C), and the specific energy (SE) are then evaluated using the following equations [8,33,34], respectively:

$$V_{av} = -\frac{E_{discharged} - E_{charged} - (x_2 - x_1)E_M}{q(x_2 - x_1)}, \quad (1)$$

$$\Delta V_{\%} = \frac{|V_{discharged} - V_{charged}|}{\min(V_{discharged}, V_{charged})} \times 100\%, \quad (2)$$

$$C = \frac{(n \times F \times 1000)}{3600 \times M_M}, \quad (3)$$

$$SE = C \times V_{av}. \quad (4)$$

In Eqs. (1)–(4), $E_{discharged}$ and $E_{charged}$ are the total energies of the discharged electrode and the commensurate charged electrode, E_M is the energy per atom of the active metal in its pure form, x_1 and x_2 are the number of metal ions in the charged and discharged electrode, respectively, q is the charge transferred by the active metal (monovalent, divalent, etc.), $V_{discharged}$ and $V_{charged}$ are the volumes of the discharged and charged electrodes, n is the number of electron transferred, F is the Faraday's constant, and M_M is the molar mass of the discharged electrode.

Using the data processing described above we were able to increase the size of the battery database from about 5,000 instances present in the original MP battery database to 191,625 instances. This new database is available as Supplementary Information. Likewise, for the AFLOW database, we obtained about 26,000 instances for battery training. Given the significantly larger number of instances in the expanded MP database, the results presented from now on will refer to this database. Results obtained for the AFLOW database can be found in Supplementary Information. A comparison of the distribution of the data based on the 10 active metal-ions for the original MP battery database (Battery) and our new expanded database (Expanded) is shown in Fig. 2. For the expanded battery data, about 33% of instances correspond to Li-ions, followed by Mg-ion which constitutes 23% (Fig. 2). In the original battery data, Li alone makes up about 55% of the data, followed by Ca with about 10%. Na and Zn each make up about 8% of the metal-ions represented (similar representation in both data sets). The least represented metal-ion in both the expanded and original databases is Cs, making up to less than 2% and 1% of the data, respectively. A comparison of both databases in terms of the distribution of the target properties, V_{av} and $\Delta V_{\%}$, is presented in Figs. 3 and 4, respectively. Our results show that the expanded data set is more balanced in terms of average voltage and $\Delta V_{\%}$ distribution as the original data is significantly more skewed towards cathode electrodes (higher voltages) and materials with smaller $\Delta V_{\%}$.

2.2. Deep learning models

With the new database of electrodes developed and the electrode properties evaluated, we set to build and train ML models to predict the target properties of interest, V_{av} and $\Delta V_{\%}$. It should be noted that the data consists of many instances that would not work well as electrode

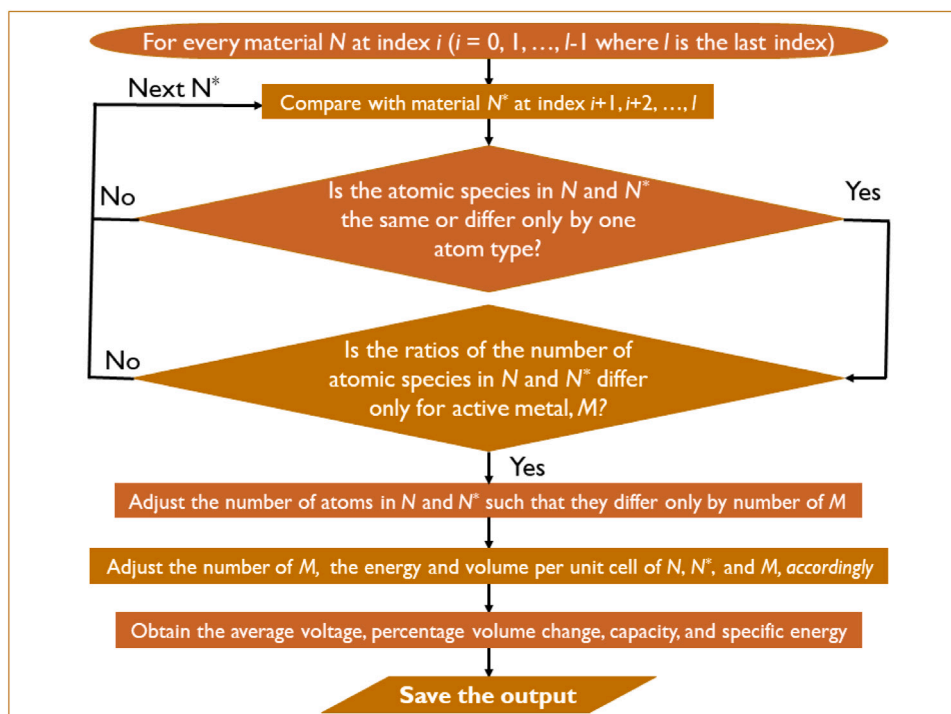


Fig. 1. Scheme of the algorithm used to search for potential electrodes.

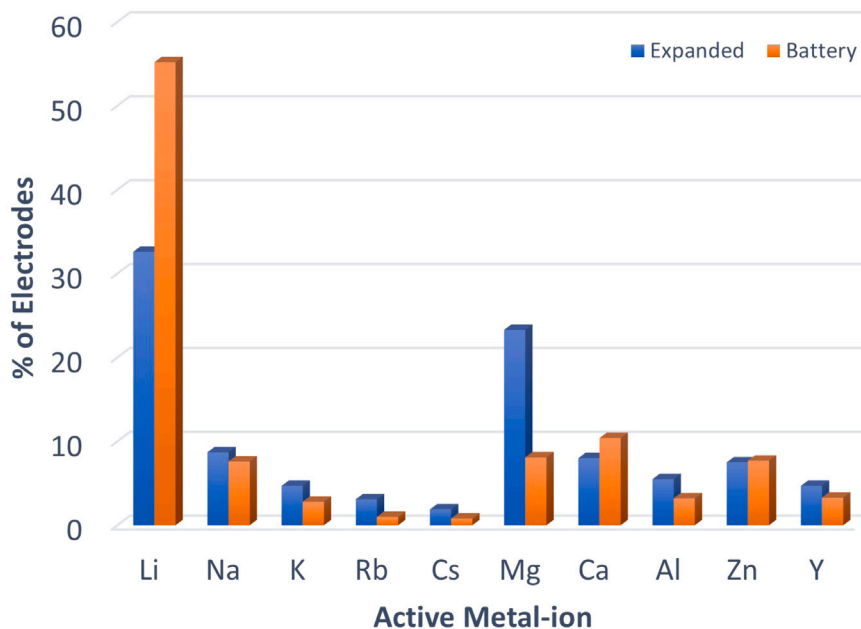


Fig. 2. Distribution of data instances in our new battery data set (Expanded) and the original battery data (Battery) from the Materials Project based on the type of active metal-ion.

materials. 17% of the data in the expanded database presents negative V_{av} corresponding to reactions that will not occur electrochemically. However, we must stress that all of these instances are greatly needed to train robust machine learning models capable of predicting if a material will work or not as metal-ion electrode in an electrochemical cell.

We then proceed to use the data to build, train, and test deep neural network models (DNN) using the Keras library [35,36]. The target properties of interest are the V_{av} and $\Delta V_{\%}$. DNN needs features, for each value of the target, that go through successive layers of representation until outputted as the predicted value of the target. As primary features we used the type of active metal, the valence of the active metal, the

number of the active metal atoms present in the charged and discharged electrodes, and the space group and crystal system of both charged and discharged electrodes (see Table 1).

The chemical formulas of the charged and discharged electrodes are then used to generate 302 other composition-based features using the matminer library [37] as described in our previous work [12]. We thus have 310 features, 4 of which are all zeros for all data instances and hence removed. The 4 features that are zero across the database are the minimum value of the number of f valence electrons, the minimum value of the number of unfilled f states, the minimum value of the bandgap, and the minimum value of the magnetic moment of

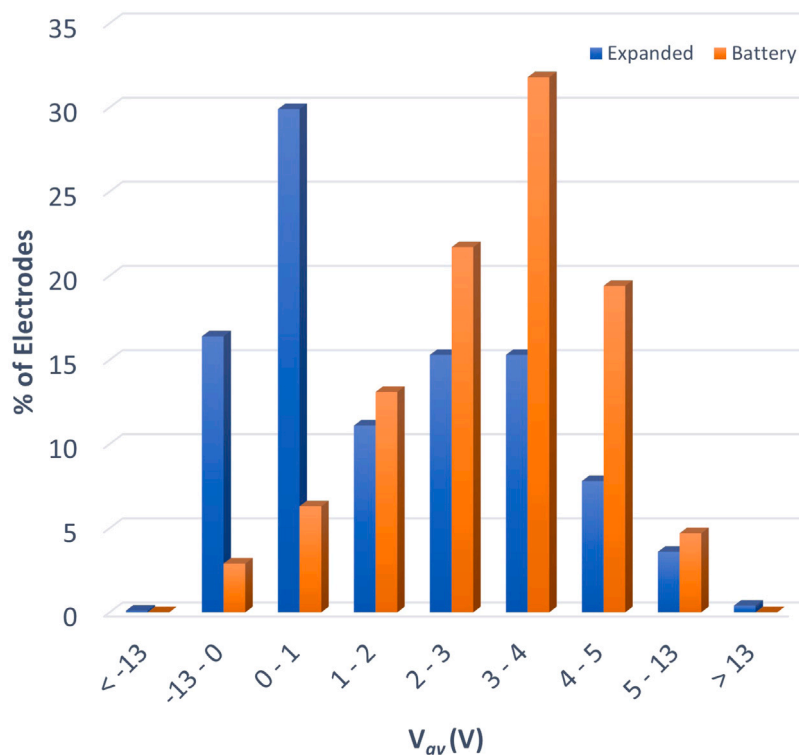


Fig. 3. Average voltage distribution of data instances in our new battery data set (Expanded) and the original battery data set (Battery) from MP.

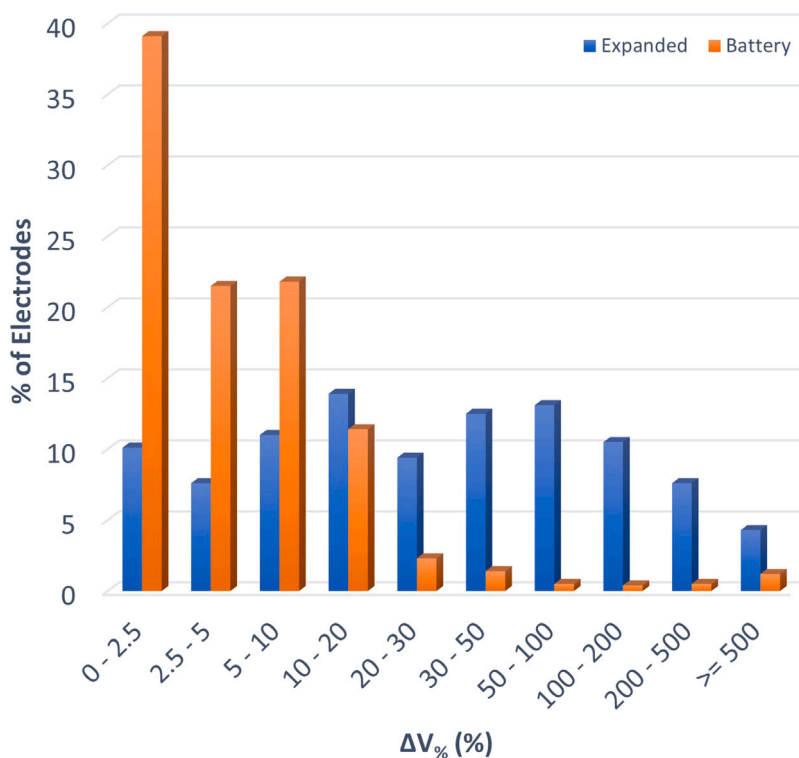


Fig. 4. Percentage volume change distribution of data instances in our new battery data set (Expanded) and the original battery (Battery) from MP.

the pure form of the atomic constituents of each of the discharged electrodes. The remaining 306 features are then normalized. In order to understand the relative relevance of these features in exploring battery electrode materials, the feature importance for V_{av} and $\Delta V_{\%}$ was determined by the permutation importance analysis (PIA) using

the random forest regressor model [38]. The first 20 most important features provided by this analysis are shown in Supporting Information. For the purpose of building the DNN, however, the principal component analysis (PCA) [39] was found to be more appropriate in obtaining the optimum feature dimension as it presents better results than the PIA

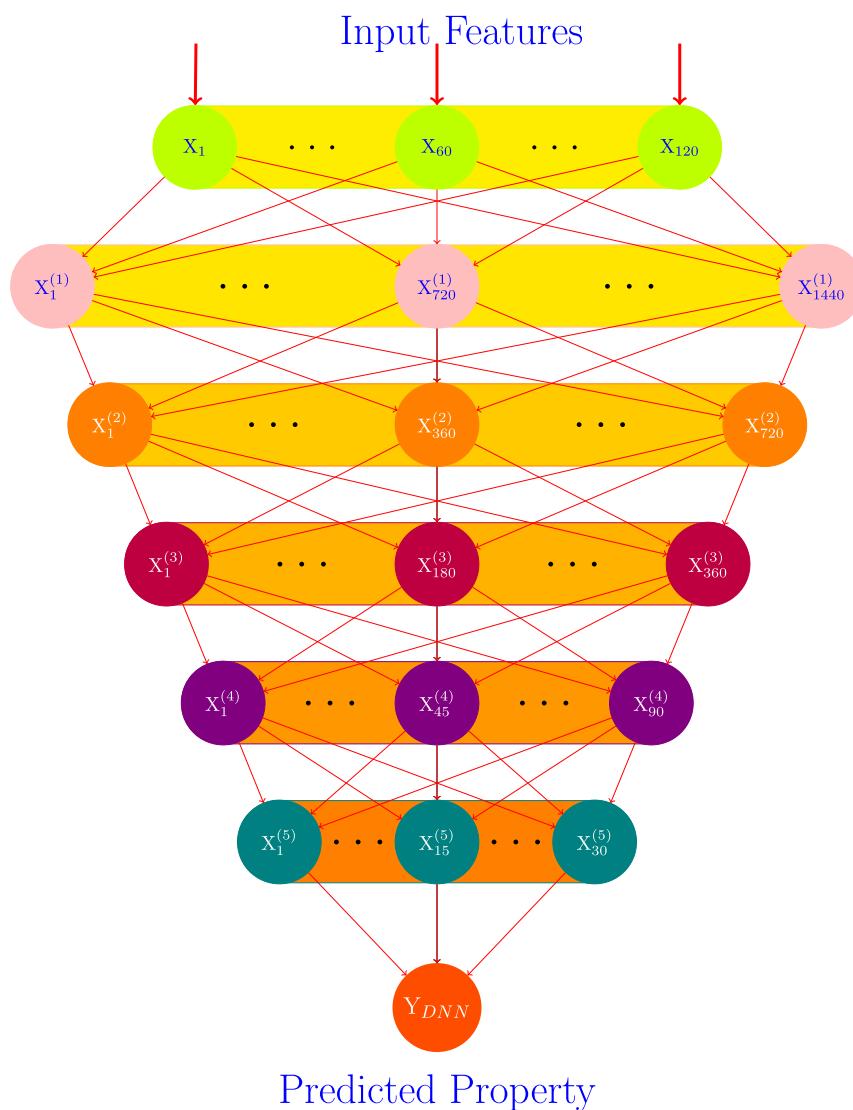


Fig. 5. The deep neural network (DNN) architecture showing the layers and their widths. The seven layers comprise the input layer (with the input features), followed by 5 hidden layers, and the output layer (predicted property).

Table 1

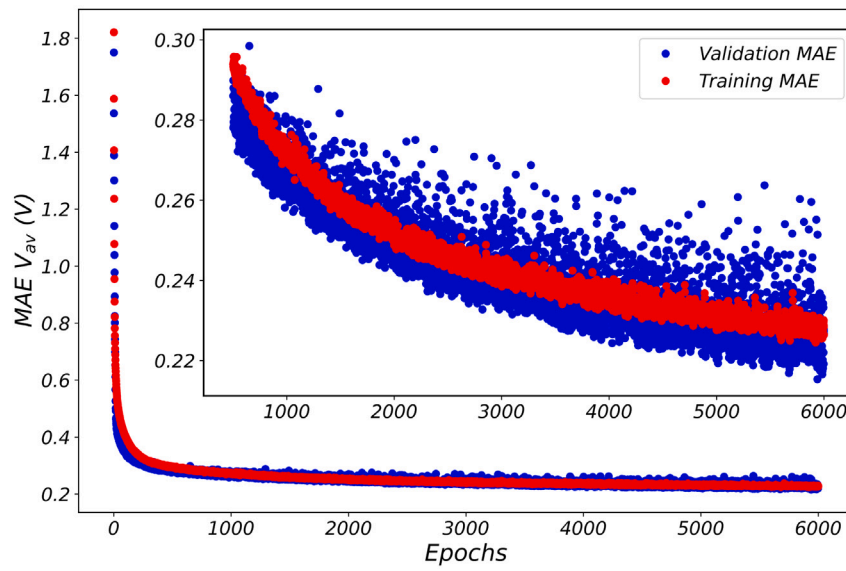
The features used in training our models. The matminer library [37] is used to generate 151 composition based features from each of the chemical formulas of charged and discharged electrodes [12].

The features	
1	The type of active metal
2	The valence of the active metal
3, 4	The number of active metal ions in charged (3) and discharged (4) electrodes
5, 6	The space group of charged (5) and discharged (6) electrodes
7, 8	The crystal system of charged (7) and discharged (8) electrodes
9, 10	The chemical formula of charged (9) and discharged (10) electrodes

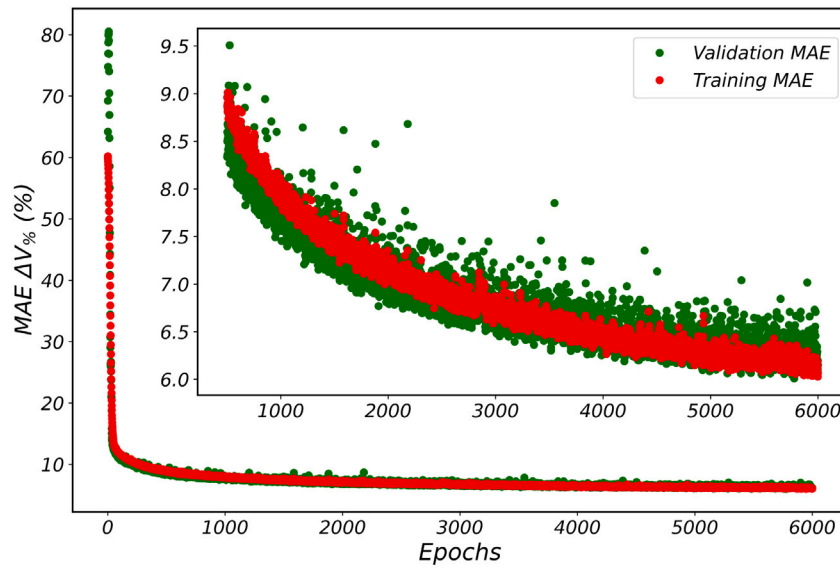
in terms of a smaller *MAE*. PCA is therefore applied to the features after which 120 features, giving about 99.8% of cumulative explained variance, were used for the training. To minimize noise from the data and to ensure our models learn well, data instances with V_{av} less than -13 V and greater than 13 V and/or $\Delta V_{\%}$ greater than 500% (amounting to about 5% of the data) were not used for building the models.

To establish the capacity of our models to predict the properties of previously reported battery data (MP original battery database), we held out about 1350 instances contained in the original battery data from the MP database. This data set (named Test data₂), is strictly reserved for testing our model exclusively on data from the original

battery database. With the Test data₂ held out for comparison purposes, we randomly selected 90% of the remaining data for training (Training data set) and the remaining 10% as additional testing data (Test data₁). These two testing data sets were chosen to test our models independently in a random sample of the newly generated expanded battery database (Test data₁) and the well established original MP battery database (Test data₂) with the goal of testing not just the accuracy of the DNN models but also the robustness of the new expanded database itself. The (hyper)parameters tuning was carried out by minimizing the *MAE* of our model with respect to the different (hyper)parameters as shown in our previous works [12,13]. The optimum model architecture showing the layers and their widths is presented in Fig. 5 while the



(a)



(b)

Fig. 6. Training progression for (a) the average voltage (V_{av}) model and (b) the percentage volume change ($\Delta V\%$) models. The inset shows a zoom-in from the epoch 500 to 6000.

(hyper)parameters used in our models are shown in Table 2. The architectures for both V_{av} and $\Delta V\%$ ML models consist of 7 layers of DNN: an input layer, 5 hidden layers, and an output layer. The activation function, the optimizer, and the loss function used in our models are the *relu*, *adam*, and *MSE*, respectively [35]. *BathNormalization*, *L2 regularization* [40], and *dropout* [41] were used to minimize over-fitting and enhance the learning while the *MAE* is chosen as the performance metric (Table 2).

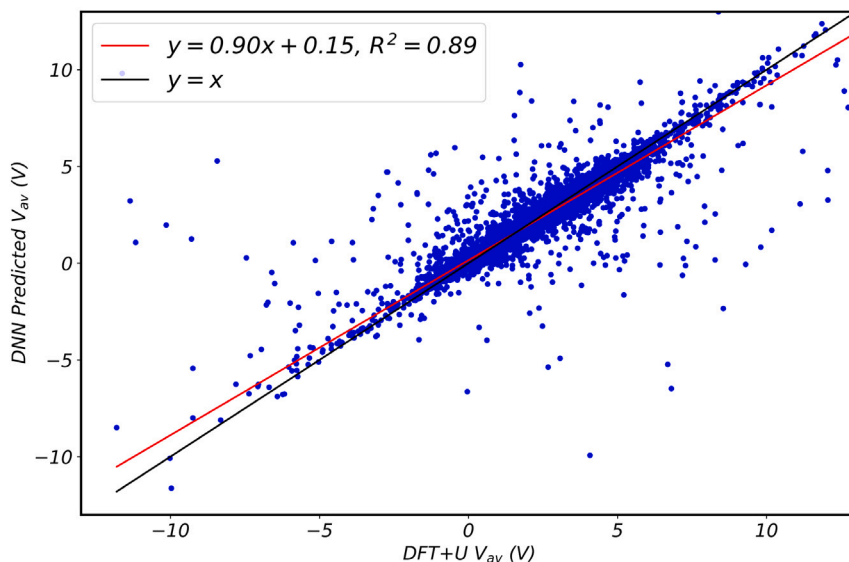
The learning progress of our models is shown in Figs. 6(a) and 6(b) for V_{av} and $\Delta V\%$, respectively. The inset in each of the figures shows a zoom-in from the epoch 500 to 6000. The cumulative minimization of the *MAE* for both the training and validation data depicts an excellent learning progression of our models that should result in a good predictive capacity. After training for about 6000 epochs, no further

minimization in the *MAE* is observed and hence the models are deemed to have optimally learned.

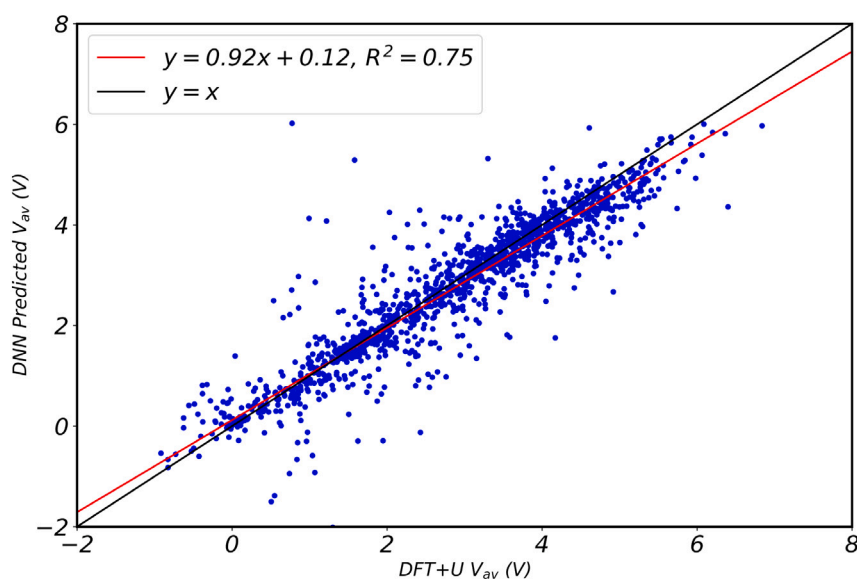
3. Results

3.1. Assessing the DNN models

We want to understand how well our DNN models can predict the target properties of electrode materials outside of the training data set. The 10% of the data set apart from testing (Test data₁) and the original battery data (Test data₂) were used to evaluate the performance of the models (Table 3). For the V_{av} prediction, our model presents a *MAE* of 0.21 V on Test data₁ with a mean absolute value (*MAV*) of 1.87 V and a *MAE* of 0.38 V on the Test data₂ with a *MAV* of 2.83 V. Our model



(a)



(b)

Fig. 7. Scatter plots for the Test data₁ (a) and Test data₂ (b) of DNN predicted values for V_{av} versus the corresponding DFT+U values.

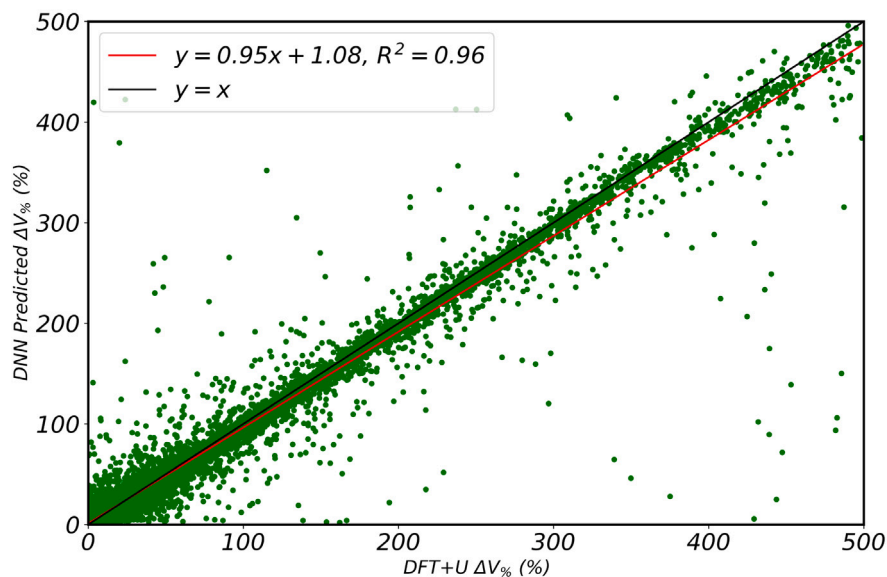
shows a good predictive capacity with average percent errors on the predicted value of V_{av} of 11% and 13% for Test data₁ and Test data₂, respectively.

To further assess the performance of our model on individual electrodes, scatter plots of the DNN predicted values against the calculated values are presented in Figs. 7(a) and 7(b). The plots show a line of best fit that deviates only slightly from the perfect fit line $y = x$. The corresponding correlation coefficients (R^2) are 0.89 and 0.75 for Test data₁ and Test data₂, respectively.

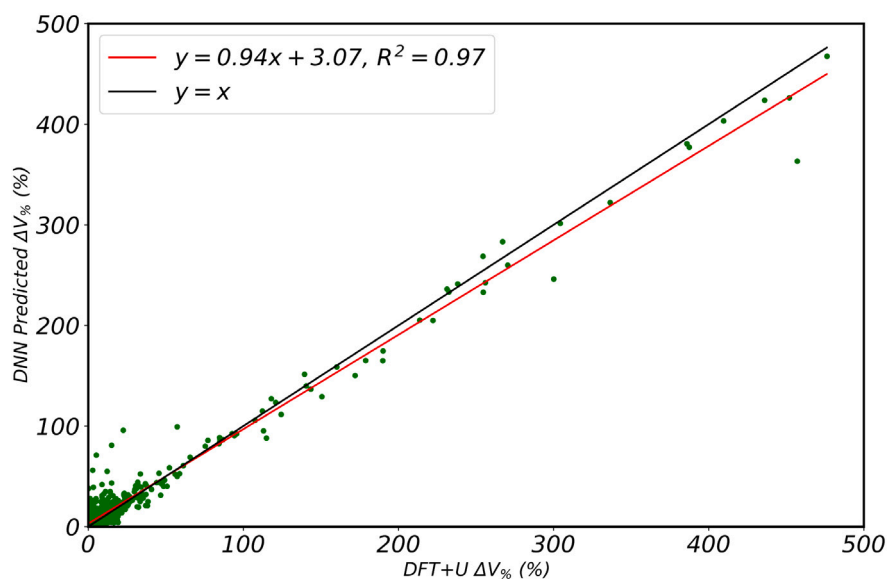
Similarly, we evaluated the performance of our models for the target $\Delta V_{\%}$. On Test data₁, we found a MAE of 6.14% while on Test data₂, the MAE is 4.54%. The scatter plots for the predicted target values against the computed values for Test data₁ and Test data₂ give R^2 values of

0.96 and 0.97 respectively, reflecting a good predictive performance on both sets (Figs. 8(a) and 8(b)).

Additionally, we selected electrodes from our Test data₂ to illustrate the performance of our models on diverse active metal-ions and electrodes with negative V_{av} and high $\Delta V_{\%}$. Table 4 shows the robustness of our models with accurate predictions for the electrode target properties. These results are particularly interesting as they show how DNN models can now effectively identify materials that would not work well as battery electrodes (negative V_{av}), as well as electrode materials that present a large volume expansion upon charging. We note that our previous ML models, trained with the original MP battery database (skewed towards cathode electrodes), were not able to accurately predict the properties of electrodes with negative or small V_{av} [12,13].



(a)



(b)

Fig. 8. Scatter plots of $\Delta V_{\%}$ predicted by DNN models versus the corresponding DFT+U values for test sets Test data₁ (a) and Test data₂ (b).

In our previous work [12], we showed how ML models can help us select a number of Na-ion electrodes with high specific energy and minimal $\Delta V_{\%}$. The Na-ion electrodes were obtained by replacing the Li atoms in Li-ion electrodes with Na ions. To further verify the robustness of our current DNN models in predicting various electrode properties, they are applied to this specific set of Na-ion electrodes identified in our previous work. The results show a quantitative improvement over the previous models built using the original MP battery data. Specifically, the MAE for V_{av} improves from 0.55 V to 0.39 V (a 29% improvement) and the MAE for $\Delta V_{\%}$ improves from 9.8% to 7.1% (about 28% improvement) (Table 5).

Recently, new ML models for V_{av} using not just composition features but also atomic coordinates have been presented [16]. These models

provide MAE improvements in V_{av} from 0.39 to 0.34 V after the 10-fold cross-validation training. By training with our expanded database using composition only models, we obtain a MAE of 0.21 V. It would be interesting to investigate if the MAE obtained in our work can be further reduced using a model such as the one presented in Ref. [16].

3.2. New battery electrodes

The expanded database does not only greatly improve the accuracy of the ML models derived from it but also presents us with a two orders of magnitude larger battery data set compared to the original MP battery database. Therefore, in addition to its potential for ML models training, this new database produces by itself electrodes that

Table 2

The (hyper)parameters used in our models. The 1st, 2nd, 3rd, 4th, and 5th, indicates the 1st, 2nd, 3rd, 4th, and 5th hidden layer, respectively.

	The (hyper)parameters	V_{av} model	$\Delta V_{\%}$ model
1	activation function	<i>relu</i>	<i>relu</i>
2	loss function	<i>mean squared error (MSE)</i>	<i>mean squared error (MSE)</i>
3	metric	<i>mean absolute error (MAE)</i>	<i>mean absolute error (MAE)</i>
4	optimizer	<i>adam</i>	<i>adam</i>
5	<i>BathNormalization</i>	Yes	Yes
6	<i>dropout</i>	0.23	0.15
7	batch size	4640	4640
8	epochs	6000	6000
9	<i>L2 regularization</i> 1st	1.2×10^{-8}	1.2×10^{-8}
10	<i>L2 regularization</i> 2nd	5.0×10^{-8}	5.0×10^{-8}
11	<i>L2 regularization</i> 3rd	5.0×10^{-7}	5.0×10^{-7}
12	<i>L2 regularization</i> 4th	5.0×10^{-6}	5.0×10^{-6}
13	<i>L2 regularization</i> 5th	6.0×10^{-5}	6.0×10^{-5}

Table 3

Data instances, mean absolute value (MAV) of the testing data sets, and mean absolute error (MAE) on testing data sets. The Train data and Test data₁ correspond to 90% and 10%, respectively, of the expanded battery data (Expanded) while the Test data₂ (Battery) is taken from the original battery data in MP. Neither set of test data were included in the training of the models.

	Train data	Test data ₁ (Expanded)	Test data ₂ (Battery)
V_{av}	163,201	18,133	1349
$\Delta V_{\%}$	163,201	18,133	1349
V_{av} MAV (V)		1.87	2.83
$\Delta V_{\%}$ MAV (%)		60.72	13.30
V_{av} MAE (V)		0.21	0.38
$\Delta V_{\%}$ MAE (%)		6.14	4.54

Table 4

The performance of the DNN models in predicting the electrode properties with wide range of target properties values and diverse active metal-ions.

Electrode				V_{av} (V)		$\Delta V_{\%}$	
	Charged	Discharged	Active Metal-ion	DFT+U	DNN	DFT+U	DNN
1	NbS ₂	KNbS ₂	K	1.61	1.66	26.6	30.0
2	NbS ₂	Li ₉ (NbS ₂) ₁₄	Li	2.39	2.51	1.3	5.5
3	NbS ₂	NaNbS ₂	Na	1.80	1.93	12.3	13.2
4	Al ₄ Cu ₁₅	AlCu ₃	Al	0.59	0.42	3.0	3.8
5	MgBi	Mg ₅ Bi	Mg	−0.01	0.02	160.2	158.8
6	TiC	Y ₃ TiC	Y	0.84	0.75	254.5	268.7
7	MnF ₄	CaMnF ₄	Ca	5.53	5.69	6.3	17.0
8	MnF ₄	MnZnF ₄	Zn	2.98	2.84	5.9	9.4
9	LiC ₁₂	LiC ₆	Li	−0.03	0.19	2.4	3.6
10	Cu ₃ O ₄	Na(Cu ₃ O ₄) ₂	Na	3.95	3.59	4.9	8.2
11	Cu ₃ O ₄	Ca(Cu ₃ O ₄) ₂	Ca	3.94	3.57	8.9	14.4
12	CaC ₄	CaC ₂	Ca	10.53	11.41	267.1	283.3
13	Al ₁₀ V	Al ₄₁ V ₄	Al	−0.02	−0.05	0.3	38.1
14	TiO ₂	Li ₃ Ti ₄ O ₈	Li	1.03	1.19	11.1	12.1
15	CrS ₂	Na(CrS ₂) ₂	Na	2.31	2.16	21.6	14.0
16	CrS ₂	K(CrS ₂) ₂	K	2.51	2.17	38.2	20.9
17	CrS ₂	LiCrS ₂	Li	2.40	2.43	11.1	16.6
18	WO ₃	AlWO ₃	Al	1.87	0.5	0.1	5.9
19	WO ₃	YWO ₃	Y	2.97	1.80	19.4	14.3
20	CuN	Ca(CuN) ₂	Ca	2.40	2.61	29.8	34.8

have not been identified or studied before. Thus, from the large data of potential battery electrodes mined, we screened out intercalation electrode materials with good performance metrics. We selected new electrodes with a specific energy of at least 1000 Wh/kg. We are also interested in materials with minimal $\Delta V_{\%}$ (less than 20%) upon charging and discharging since small $\Delta V_{\%}$ enhances the coulombic efficiency, cyclability and safety of the battery [42,43]. Since the cell voltage stability window of most electrolytes is less than 5 V [44], cathodes with greater than 7 V of V_{av} are also excluded. We found about 35 novel electrodes for different active metals; Li, Na, Mg, Ca, and Al. From their crystal structure, we observe that these electrodes present clear ion diffusion pathways and specific energies that can be as high as 3000 Wh/kg for MnO₄ and Fe(O₂F)₂ (Table 6). These materials, to

Table 5

The performance of the DNN models in predicting the electrode properties of Na-ion electrodes. The Na-ion electrodes are obtained by replacing the Li in Li-ion electrodes with Na. The performance shows significant improvement of our DNN built on the expanded battery data over that built on only battery data. Calc. values are calculated using DFT+U method.

Reaction		$\Delta V_{\%}$			V_{av} (V)		
		DNN	ML ^a	Calc. ^a	DNN	ML ^a	Calc. ^a
1	NaMn ₃ OF ₈ $\leftrightarrow$ Na ₄ Mn ₃ OF ₈	13.3	5.6	26.0	3.45	3.2	3.15
2	Na(CoO ₂) ₂ $\leftrightarrow$ Na ₂ CoO ₂	60.3	4.1	33.2	1.64	2.74	1.24
3	NaNbTe ₂ WO ₁₂ $\leftrightarrow$ Na ₄ NbTe ₂ WO ₁₂	15.9	4.3	15.2	3.14	3.25	3.1
4	NaCuF ₄ $\leftrightarrow$ Na ₂ CuF ₄	18.5	11.8	20.5	4.40	4.01	4.31
5	Na ₂ CrO ₄ $\leftrightarrow$ Na ₄ CrO ₄	2.4	5.6	2.2	1.50	3.07	1.46
6	NaMn ₂ F ₉ $\leftrightarrow$ Na ₃ Mn ₂ F ₉	14.5	7.1	14.2	5.03	3.67	4.35
7	Mn ₃ P ₆ WO ₂₄ $\leftrightarrow$ Na ₆ Mn ₃ P ₆ WO ₂₄	3.5	6.9	25.5	3.63	3.50	3.47
8	CrWO ₆ $\leftrightarrow$ Na ₂ CrWO ₆	4.8	4.9	3.5	4.15	3.10	3.75
9	NaNbF ₆ $\leftrightarrow$ Na ₃ NbF ₆	11.2	4.1	1.6	1.38	2.56	0.83
10	NaVTe(WO ₆) ₂ $\leftrightarrow$ Na ₄ VTe(WO ₆) ₂	12.9	4.5	13.8	3.23	3.20	2.90
11	NaSbF ₆ $\leftrightarrow$ Na ₃ SbF ₆	31.4	11.3	15.5	2.11	3.79	2.07
12	NaVOF ₁₁ $\leftrightarrow$ Na ₄ VOF ₁₁	5.5	3.7	1.7	5.36	3.83	3.50
13	NaV ₂ O ₄ $\leftrightarrow$ Na ₂ VO ₂	27.2	3.3	42.3	0.50	2.85	−0.35
14	NaV ₂ O ₇ $\leftrightarrow$ Na ₃ V ₂ O ₇	5.9	3.4	19.5	3.12	3.76	3.11
15	CrPO ₄ F $\leftrightarrow$ Na ₃ Cr ₂ P ₂ (O ₄ F) ₂	20.5	4.1	22.1	3.91	3.75	3.26
16	NbO ₂ F $\leftrightarrow$ NaNbO ₂ F	7.2	3.1	3.2	1.46	3.31	1.36
17	TiCrO ₄ $\leftrightarrow$ Na ₂ TiCrO ₄	23.4	2.4	27.0	2.06	3.31	1.54
18	NaTiV ₃ O ₁₀ $\leftrightarrow$ Na ₄ TiV ₃ O ₁₀	3.8	2.4	1.1	2.73	3.00	2.47
19	Mn ₂ CrO ₆ $\leftrightarrow$ Na ₃ Mn ₂ CrO ₆	20.4	2.9	25.0	2.85	2.95	2.73
20	NaMn(PO ₄) ₂ $\leftrightarrow$ Na ₃ Mn(PO ₄) ₂	25.3	3.0	23.6	3.96	3.28	4.32
21	NaMn ₈ O ₁₆ $\leftrightarrow$ Na ₃ Mn ₄ O ₈	14.5	3.6	24.1	2.49	2.99	1.98
22	VF ₅ $\leftrightarrow$ Na ₂ VF ₅	7.7	6.4	11.9	4.38	3.61	4.01
MAE		7.1	9.8		0.39	0.55	

The ML^a and Calc.^a are taken from the Ref. [12].

the best of our knowledge, have not been previously studied as metal-ion electrodes. It should be stressed, however, that further assessment of intermediate states of charge must be performed using quantum mechanical methods and ultimately experiments to confirm their full potential as electrochemical electrodes.

4. Conclusions

We present an efficient screening of hundreds of thousands of inorganic materials from the Materials Project and the AFLOW data sets to create an expanded database for metal-ion electrode materials for rechargeable batteries. Our work provides an expanded and more balanced battery database for 10 active metal ions that contains about 190,000 instances, a two order of magnitude increase with respect to the original MP battery data set. This data is made available as Supplementary Information. The new data set is far less skewed towards cathode materials than the original MP battery data set and incorporates a significant number of *bad* electrodes with negative voltages. The expanded battery data set includes instances with computed average voltage (V_{av}), percentage volume change ($\Delta V_{\%}$) upon charging and discharging, specific capacity, and specific energy for each pair of charged and discharged electrodes. From the new database itself, we identified 35 novel intercalation electrodes with excellent performance in terms of $\Delta V_{\%}$ and specific energy. In addition, the large and diverse battery data was used to build regression-based deep neural network models. Testing our models on testing data instances which were never included in the training reveals the excellent performance of our DNN models. The V_{av} model gives a mean absolute error (MAE) of 0.21 V on a testing data set with a 1.87 V of mean absolute value (MAV), and a MAE of 0.38 V on an additional testing data set with a 2.83 V MAV (original MP battery database). Likewise, the $\Delta V_{\%}$ model gives a MAE of 6.14% on a testing database with a 60.72% MAV, and a MAE of 4.54% on a testing data set with a 13.30% MAV. Further testing confirmed the robustness of our models as they outperformed by at least 28% our previous models (for both the V_{av} and $\Delta V_{\%}$) trained only with the available battery data from MP, and underscores the critical role of a robust data set in building accurate DNN models.

Table 6

Novel electrodes generated solely by screening the new expanded database under the conditions of presenting less than 20% $\Delta V_{\%}$, less or equal to 7 V of V_{av} , and a specific energy (SE) greater than 1000 Wh/kg. Sym represents the space group number of the electrode and SE is the specific energy.

S/No.	Discharged	Sym	Charged	Sym	ion	V_{av} (V)	$\Delta V_{\%}$ (%)	C (mAh/g)	SE (Wh/kg)
1	Li ₂ MnO ₄	31	MnO ₄	141	Li	3.5	17.2	872.3	3055.5
2	Li ₇ Fe(O ₂ F) ₂	1	Fe(O ₂ F) ₂	29	Li	3.3	18.5	908.9	3002.1
3	Li ₃ TiS ₃	15	TiS ₃	160	Li	2.5	11.4	487.7	1208.0
4	Li ₃ RhF ₆	225	RhF ₆	229	Li	4.4	10.0	338.2	1491.5
5	Li ₆ Cr ₂ O ₇	15	Li ₂ Cr ₂ O ₇	14	Li	2.8	10.5	416.1	1174.7
6	Li ₇ Fe ₃ O ₁₀	2	Li ₂ Fe ₃ O ₁₀	2	Li	3.6	3.1	356.3	1266.5
7	K ₈ Li ₁₃ (FeO ₄) ₄	1	K ₂ FeO ₄	62	Li	2.6	11.8	394.9	1027.1
8	Li ₄ Mn ₃ NiO ₈	8	Mn ₃ O ₈	9	Li	3.8	9.1	297.3	1144.7
9	Li ₄ CuF ₅	87	Li ₂ CuF ₅	14	Li	3.9	3.4	287.7	1129.8
10	Li ₂ Si ₂ NiO ₆	7	Si ₂ NiO ₆	1	Li	4.7	2.7	238.5	1124.7
11	Li ₃ IO ₆	151	Li ₂ IO ₆	169	Li	4.8	11.8	312.1	1500.7
12	Li ₂ CuSiO ₄	14	CuSiO ₄	148	Li	4.3	7.6	316.2	1360.4
13	Li ₃ OsO ₆	1	Li ₂ OsO ₆	15	Li	3.0	11.5	400.3	1208.2
14	Li ₃ CuPO ₅	33	CuPO ₅	19	Li	5.3	17.1	411.6	2198.3
15	LiCrSiO ₄	95	CrSiO ₄	230	Li	6.0	10.8	177.5	1070.8
16	LiSiNiO ₄	74	SiNiO ₄	230	Li	6.5	0.1	169.9	1106.7
17	Li ₃ Cu ₂ F ₇	14	LiCu ₂ F ₇	14	Li	5.4	10.0	190.8	1030.2
18	Li ₂ FeF ₅	63	LiFeF ₅	88	Li	6.4	11.2	162.7	1036.4
19	Li ₃ Ti ₆ FeO ₁₆	4	Ti ₆ FeO ₁₆	1	Li	5.1	11.3	211.5	1081.9
20	Na ₃ RhF ₆	225	RhF ₆	229	Na	4.6	8.3	281.3	1282.7
21	Na ₂ FePO ₄ F	60	FePO ₄ F	52	Na	4.8	0.8	248.4	1192.4
22	MgZrN ₂	141	ZrN ₂	164	Mg	2.8	6.9	373.4	1063.6
23	MgMoO ₄	12	MoO ₄	2	Mg	4.6	1.2	290.9	1326.5
24	MgRhF ₆	148	RhF ₆	62	Mg	4.7	3.3	222.2	1038.1
25	Mg ₃ Mo ₂ O ₇	36	Mo ₂ O ₇	11	Mg	2.8	0.7	426.8	1181.8
26	Mg ₃ Fe ₂ O ₅	12	MgFe ₂ O ₅	63	Mg	2.6	7.5	405.2	1042.4
27	MgCrSiO ₅	15	CrSiO ₅	9	Mg	3.6	10.5	290.7	1033.3
28	MgFeF ₆	148	FeF ₆	15	Mg	5.4	15.1	276.1	1486.2
29	Ca ₃ Cu ₂ (ClO ₂) ₂	139	CuClO ₂	5	Ca	3.2	9.2	420.7	1347.6
30	LiCaVF ₆	163	LiVF ₆	4	Ca	4.4	10.2	252.9	1101.0
31	Ca ₄ PtO ₆	167	CaPtO ₆	15	Ca	3.3	6.3	356.3	1159.1
32	Ca ₂ MgWO ₆	14	MgWO ₆	14	Ca	3.6	7.2	279.0	1013.5
33	CaFeF ₆	148	FeF ₆	15	Ca	6.1	4.7	255.4	1551.4
34	CaCrF ₆	148	CrF ₆	148	Ca	5.9	10.6	260.1	1546.0
35	AlVO ₃	148	VO ₃	74	Al	2.7	0.9	638.5	1710.3

CRedit authorship contribution statement

Isaiah A. Moses: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Veronica Barone:** Conceptualization, Methodology, Resources, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration. **Juan E. Peralta:** Validation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Database available as Supplementary Information.

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Appendix A. Supplementary data

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